

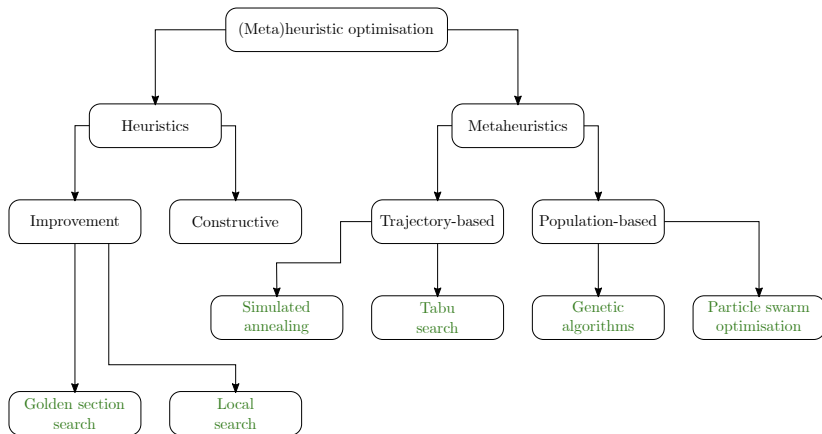
GENETIC ALGORITHMS

Lecture 25: Presented by Jacomine Grobler



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(Meta)heuristics



In trajectory-based metaheuristics, a single solution is maintained and updated iteratively over time.

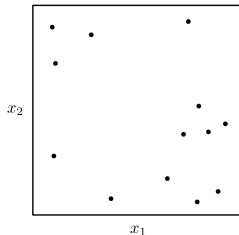
In population-based metaheuristics, an entire population containing multiple candidate solutions is maintained simultaneously and updated iteratively over time.

Genetic algorithms are a class of population-based metaheuristics based on the theory of evolution by natural selection proposed by Charles Darwin.

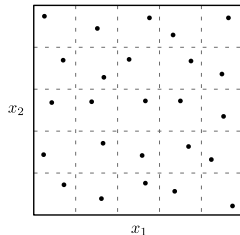
Components of a genetic algorithm:

- Population initialisation (comprising multiple individuals)
- Fitness function
- Parent selection
- Crossover
- Mutation
- Replacement scheme / survivor selection
- Termination criterion

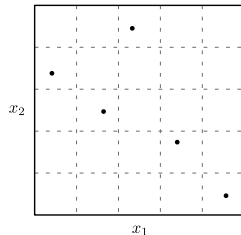
Population initialisation



(a) Random generation



(b) Parallel diversification



(c) Sequential diversification

- Random generation
- Parallel diversification
- Sequential diversification
- Hybrid initialisation
- Heuristic initialisation

Diversity *vs* computational burden *vs* solution quality

Maximisation:

- Use the objective function $f(\mathbf{x})$ as a proxy for the fitness function of an individual $F(\mathbf{x})$

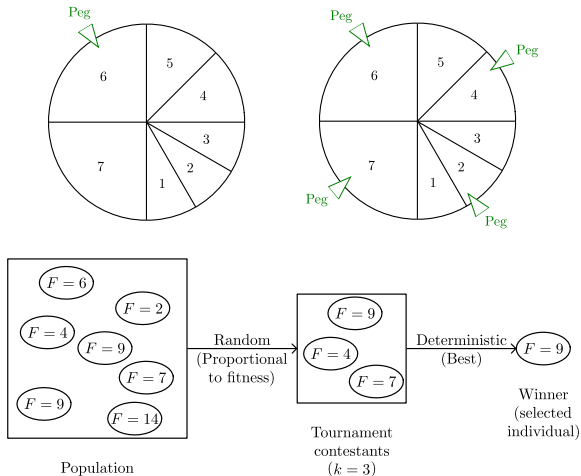
Minimisation:

- The fitness function value $F(\mathbf{x})$ of any individual $\mathbf{x}$ should be positive
- If $f(\mathbf{x})$ is non-negative on $\mathbb{R}^n$: $F(\mathbf{x}) = 1/f(\mathbf{x})$
- Else: $F(\mathbf{x}) = U - f(\mathbf{x})$

Where U is a known upper bound on $f(\mathbf{x})$.

Parent selection strategy

- Roulette wheel selection
- Stochastic universal sampling
- Tournament selection



In the application of these crossovers, two parents

$\mathbf{x}^1 = [x_1^1 \ \cdots \ x_n^1]^T$ and $\mathbf{x}^2 = [x_1^2 \ \cdots \ x_n^2]^T$ produce offspring

$\mathbf{y}^1 = [y_1^1 \ \cdots \ y_n^1]^T$ and $\mathbf{y}^2 = [y_1^2 \ \cdots \ y_n^2]^T$, where:

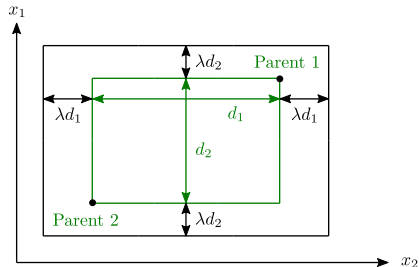
- Intermediate crossover

$$y_i^1 = \alpha^1 x_i^1 + (1 - \alpha^1) x_i^2 \quad \text{and} \quad y_i^2 = \alpha^2 x_i^1 + (1 - \alpha^2) x_i^2$$

- Geometric crossover

$$y_i^1 = (x_i^1 x_i^2)^{\alpha^1} \quad \text{and} \quad y_i^2 = (x_i^1 x_i^2)^{\alpha^2}$$

- Simplex crossover



- Simulated binary crossover

Generate n random numbers $\mu^1, \dots, \mu^n$ uniformly within the unit interval $[0, 1]$. Thereafter,

$$\beta^i = \begin{cases} 2(\mu^i)^{\frac{1}{1+\eta}} & \text{if } \mu^i \leq \frac{1}{2} \\ \left(\frac{1}{2(1-\mu^i)}\right)^{\frac{1}{1+\eta}} & \text{otherwise.} \end{cases}$$

Using the values $\beta^1, \dots, \beta^n$, two offspring $\mathbf{y}^1 = [y_1^1 \ \dots \ y_n^1]^T$ and $\mathbf{y}^2 = [y_1^2 \ \dots \ y_n^2]^T$ are generated with i -th coordinates

$$\begin{aligned} y_i^1 &= \frac{1}{2} [(1 + \beta^i)x_i^1 + (1 - \beta^i)x_i^2] \\ y_i^2 &= \frac{1}{2} [(1 - \beta^i)x_i^1 + (1 + \beta^i)x_i^2] \end{aligned}$$

from two parents $\mathbf{x}^1 = [x_1^1 \ \dots \ x_n^1]^T$ and $\mathbf{x}^2 = [x_1^2 \ \dots \ x_n^2]^T$.

- Parent centric crossover

This operator is applied by computing the population centroid vector $\mathbf{c}$. Thereafter, direction vectors $\mathbf{d}^1 = \mathbf{x}^1 - \mathbf{c}$ and $\mathbf{d}^2 = \mathbf{x}^2 - \mathbf{c}$ are computed from $\mathbf{c}$ to the two parents $\mathbf{x}^1 = [x_1^1 \ \cdots \ x_n^1]^T$ and $\mathbf{x}^2 = [x_1^2 \ \cdots \ x_n^2]^T$ producing offspring. Upon taking $\mathbf{e}^i$ to be a unit vector orthogonal to $\mathbf{d}^i$ for $i = 1, 2$, the offspring are computed as

$$\mathbf{y}^i = \mathbf{x}^i + \xi^i \mathbf{d}^i + \zeta^i \|\mathbf{x}^1 - \mathbf{x}^2\|_2 \mathbf{e}^i, \quad i = 1, 2,$$

where ξ^i is a random number sampled from a normal distribution with zero mean and variance

$\sigma_\xi^2 = \frac{1}{2}[(\|\mathbf{d}^1\|_2 - d)^2 + (\|\mathbf{d}^2\|_2 - d)^2]$, with

$d = (\|\mathbf{d}^1\|_2 + \|\mathbf{d}^2\|_2)/2$, and where ζ^i is a random number sampled from a normal distribution with zero mean and variance $\sigma_\zeta^2 = \|\mathbf{x}^1 - \mathbf{x}^2\|_2^2$.

A pre-specified probability p_m determines the likelihood of mutating each element (gene) of a small number of individuals selected randomly to undergo mutation.

Small values are recommended for the probability p_m so that, on average, only one component value is altered per individual that undergoes mutation.

Two criteria:

- *Ergodicity*: The mutation operator should allow every solution of the search space to be reachable.
- *Locality*: The operator should produce a minimal change (preferably resulting in an individual in the local neighbourhood of the original individual).

- Generational replacement:
Offspring population replaces entire parent population
- Steady-state replacement:
Only two offspring are generated, which replace the two least fit individuals of the parent population.

- Static termination criteria
- Dynamic or adaptive termination criteria

Algorithm 4.6 (Genetic algorithm metaheuristic)

Input: A multivariate function $f : \mathbb{R}^n \mapsto \mathbb{R}$, a population size N , a selection strategy, a crossover operator, a mutation operator, a replacement strategy, and a termination criterion

Output: An approximation of a local optimum of the function f

Step 1 Generate an initial population $\mathcal{P}_0$ of candidate solutions stochastically

Step 2 $t \leftarrow 0$ and $\mathbf{x}^* \leftarrow$ the fittest member of $\mathcal{P}_0$

Step 3 $\mathcal{P}'_t \leftarrow \text{Selection}(\mathcal{P}_t)$

Step 4 $\mathcal{P}'_t \leftarrow \text{Crossover}(\mathcal{P}'_t)$

Step 5 $\mathcal{P}'_t \leftarrow \text{Mutation}(\mathcal{P}'_t)$

Step 6 $\mathcal{P}_{t+1} \leftarrow \text{Replacement}(\mathcal{P}_t, \mathcal{P}'_t)$

Step 7 If $\mathcal{P}_{t+1}$ contains a solution $\mathbf{x}$ that is better than $\mathbf{x}^*$, then $\mathbf{x}^* \leftarrow \mathbf{x}$

Step 8 $t \leftarrow t + 1$

Step 9 If the termination criterion is satisfied, output $\mathbf{x}^*$ and stop

Step 10 Return to Step 3

Example 4.7 (pp. 144–146)

Minimise the function

$$f_{4.3}(x_1, x_2) = (x_1 - 3.14)^2 + (x_2 - 2.72)^2 + \sin(3x_1 + 1.41) + \sin(4x_2 - 1.73)$$

approximately on the domain $(x_1, x_2) \in [0, 6] \times [0, 6]$ by applying five iterations of a genetic algorithm.

